

# Márk Gyulai

## Senior Java Developer

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## About me

I develop backend systems based on Java and Kotlin, with a particular focus on sustainable architecture, maintainability and long-term evolvability. As an AI consultant, I help companies identify the areas where artificial intelligence can create real business value, and I support its practical, goal-oriented adoption.

## Experience

### Adroit Group Senior Software Engineer

October 2025 – present

I have taken part in three projects, two of them in Kotlin. I develop backends, take part in design and pay attention to quality. One of the projects is AI consulting.

- **MOL Limo rent a car** (Kotlin, Spring Boot, Spring Security, PostgreSQL, Exposed, Jenkins, GitLab, Jira, Confluence, Claude): I develop the backend of the MOL Limo car rental system in Kotlin.
  - Business logic and calculations in the backend, beyond aggregating data
  - Asynchronous synchronization with a subsystem
  - Managing a PostgreSQL database, including transaction management
  - Implementing a custom OAuth flow and token handling with Spring Security
  - Scheduled jobs with Spring's @Scheduled annotation
  - Documentation-first (design-first) approach
  - Writing unit and integration tests, and manual testing
  - Communicating with the client
- **Helvo** (Kotlin, Spring Boot 4, OpenAPI, GitLab, ArgoCD, Jira, Confluence, Claude): My international team and I built a backend for frontend application for a Swiss fintech startup in Kotlin with Spring Boot 4.
  - Documentation-first (design-first) approach: designing the API in OpenAPI
  - Connecting the backend to other subsystems and the frontend with artifacts generated from the OpenAPI description
- **GLI Solutions – AI Consultant and Developer** (C#, .NET, Node.js, Angular, PostgreSQL, AI, MCP, Prompting, Azure DevOps, Jira, Confluence, Claude): As an AI consultant, I look into where artificial intelligence is useful, whether it is feasible and how it fits into existing systems. My team and I also work on the project as developers.
  - Developing a platform that supports manual testing, with a C# and .NET backend, a PostgreSQL database, Node.js and Angular
  - The platform can be connected to AI agents as an MCP server
  - Transaction management in the PostgreSQL database
  - Code-first approach
  - Handling bugs
  - Steering the team and the code towards clean code
  - Implementing repository-level AI skills
  - Writing documentation

Plane

### RDI Software Hungary Software Engineer

August 2024 – October 2025

I worked on the McDonald's application, in the team responsible for user accounts, authentication and authorization. My team and I developed and maintained microservices, together with their unit and integration tests. I wrote SQL queries and connected other systems' APIs. Unofficially, I mentored a junior colleague here too.

- **McDonald's** (Java, Microservices, REST API, OpenAPI, SQL, NoSQL, Python, Authentication, Authorization, Unit tests, GitHub, ArgoCD, Jira, Confluence, Copilot): User accounts, authentication and authorization; I worked in an international team.
  - Maintaining existing microservices
  - Designing and developing new microservices
  - Designing API endpoints and tidying up the OpenAPI description file with reusable objects
  - Finding and fixing bugs based on metrics, traces and logs
  - Raising test coverage above 90%
  - Repairing data corrupted by migrations in a NoSQL database with a Python script
  - Managing database transactions
  - Code reviews
  - Google login with the Google SDK instead of plain API endpoints

### EPAM Systems Software Engineer
January 2022 – August 2024

As part of an international team, we developed the financial application of Broadridge, an American company. This is where I got to know multithreaded programming, domain-driven design and the functional approach in depth. I wrote unit, system and acceptance tests, as well as UI tests with Selenium. I connected other systems' APIs, sometimes asynchronously. I fixed concurrency bugs, handled database transactions and improved test coverage. I also mentored a junior developer.

- **Broadridge financial application** (Java, Multithreading, DDD, Selenium, SQL, Jenkins, GitLab, Jira, Confluence)
- **Multitenant framework** (Java, Multitenancy, Jenkins, GitLab, Jira, Confluence, Copilot): The multitenant framework of the Broadridge project, which I worked on as a separate project.

### EPAM Systems Junior Software Engineer
July 2021 – January 2022

As a junior developer I built backend features for EPAM's internal exam system, and later worked on the Broadridge project as well.

- **Internal exam system** (Java, Spring Boot 3, MySQL, JUnit, Unit tests, Jenkins, GitLab, Jira, Confluence): EPAM's internal exam system: a web application in which exams could be taken safely. My team and I built its backend with Spring Boot 3 and a MySQL database.
  - Designing and developing new features
  - Transaction management in the MySQL database
  - Code reviews
  - Writing unit tests with JUnit
  - Manual testing
  - Learning new technologies and developing my skills
- **Broadridge financial application** (Java, Jenkins, GitLab, Jira, Confluence): I later worked on the same project as a Software Engineer.

## Skills

AI and Model Context Protocol (MCP) · Java 8–25 · Kotlin · Spring Boot · Software design and development principles · Testing · Tools · Angular

## Side projects

**lizacover.hu**

**This portfolio** github.com/gyulaimrk/website

## Education and certificates

**Mathematics and Computer Science, BSc**
University of Miskolc, 2017 – 2021

**Java Mentored Education Program**
EPAM Systems, September 2020 – July 2021

**English language exam, B2**

**Languages**
Hungarian, English (B2 language exam)

**Driving licence**
Category B

### Anthropic certificates

**Building with the Claude API**
Anthropic, March 2026

**Introduction to Model Context Protocol**
Anthropic, March 2026

**Introduction to agent skills**
Anthropic, March 2026

**Claude Code in Action**
Anthropic, March 2026

### More certificates (LinkedIn Learning)

**Microservices Foundations** · June 2021
**Docker for Developers** · January 2024
**Introduction to Docker for Java Developers** · January 2024
**Java 17: First Look** · January 2024
**Java: Automated API Testing with REST Assured** · January 2024
**Learning Java 17** · October 2023
**Java 8 for Professionals** · October 2023
**First Look: Java 10 and Java 11** · October 2020